PEN FOUNDATION

Website Design & Content Brief

C-DISC Technologies Pvt. Ltd. · Kozhikode, Kerala

Document Type	**Version**
Website brief for web development company. Covers all pages, sections, interactions, content, and CTAs.	v1.0 — March 2026. Phase 1 scope only.

CONTEXT

About PEN Foundation

What PEN Foundation Is

PEN Foundation — Pre-Engineered Nail Foundation

Biomimetic. Rapid Deploy. Zero Excavation.

PEN Foundation is a patented foundation technology that deploys rapidly and bears structural load immediately — no curing, no waiting, no concrete cast on site. Developed by C-DISC Technologies Pvt. Ltd. and validated by NIT Calicut and IIT Kanpur, it replaces the conventional cast-in-situ footing, eliminating the 21-day curing cycle that has defined foundation construction for over a century. The moment the last nail is driven, the green signal is given — structure development begins the same day.

The system draws its load-transfer logic from nature. The same way a tree anchors against wind through distributed root friction, PEN transfers structural load through four battered galvanized steel nails driven into the surrounding soil. This creates a three-dimensional soil-structure interaction zone that engages a significantly larger soil volume than a conventional pad footing — delivering a field-validated Enhancement Factor of 1.5–3× over conventional foundations across varied soil conditions, confirmed through field testing at NIT Calicut and parametric modelling across multiple soil types.

Sustainability is built into the mechanism itself, not added on after. PEN uses over 80% less concrete than a conventional isolated footing. Zero soil is excavated — eliminating 2–3 m^3 of spoil disposal per foundation point while preserving natural soil stratification, groundwater flow, and aquifer recharge. Tungsten carbide penetration tips are sourced from repurposed industrial drill bits, closing the materials loop before the foundation is even in the ground. At end of life, components are recoverable and reusable. A full Life Cycle Assessment has been conducted per ISO 14040/14044.

Patented. GRIHA Certified. Field-tested by NABL-accredited MatterLab. Validated by NIT Calicut and mentored by IIT Kanpur. A dedicated BIS standard is currently under development.

[OR]

PEN Foundation is a patented, sustainability-first alternative to conventional footings that installs rapidly and bears load immediately — no excavation, no wet concrete on site, no 21-day curing wait. Where a conventional footing sits idle for three weeks, PEN is already done: a biomimetic load-transfer mechanism, inspired by the way tree roots anchor into soil, delivers field-validated capacity of 1.5–3× conventional foundations — confirmed by NIT Calicut and IIT Kanpur. Over 80% less concrete. Zero excavated spoil. Recycled tungsten carbide tips. Fully recoverable at end of life. GRIHA Certified. Patented.

The Three Core Claims

> These three claims must appear on every page of the website in some form. They are the product's entire value proposition compressed into three lines. Do not alter the wording.

Claim	Conventional	PEN Foundation
Installation time	21+ days (includes curing)	~2 hours. No curing. Immediate load-bearing.
Load capacity	Dependent on soil bearing capacity at base	2.0–2.6× SBC — Enhancement Factor field-validated at NIT Calicut
Excavation	Required — soil displaced, groundwater disrupted	Zero excavation. Zero groundwater disruption. Soil integrity preserved.

Technical Specifications

These figures are validated and must appear verbatim on the website wherever technical data is displayed. The development team must not paraphrase, estimate, or substitute these values.

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MODEL             CD-PEN-32.3.2.1500
GOVERNING CODE    IS 2911 Part 4 (Nail / Pile System)
FACTOR OF SAFETY  2.5 (one plate load test)  /  2.0 (two or more
tests)
NODE MATERIAL     M50 Grade Precast Concrete  —  450 × 450 × 200mm
NAIL PIPES        GI Pipe  —  32mm OD, 3.2mm wall, IS 1239
NAIL TIP          Tungsten Carbide (recycled Kennametal / Sandvik
grade)
BATTER ANGLE      40–51°  —  site-adjusted
MIN EMBEDMENT     900mm  —  per IS 2911 Part 4
GROUT             Fosroc Conbextra GP2 micro-concrete, non-shrink
BOLTS             M12 Grade 4.6 Galvanised  —  IS 5624
PEAK LOAD OBS.    667 kN/m²  (MatterLab field test, Wayanad)
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Validation & Certifications

Every validation listed below must appear on the website. These are the proof points that convert skeptical structural engineers and institutional buyers.

Institution / Body	What Was Validated	Output
NIT Calicut	Field plate load tests per IS 1888:1982	Enhancement Factor 2.0–2.6× confirmed

IIT Kanpur — Dr. Nihar Ranjan Patra	Mentorship & long-term monitoring programme	Mentorship agreement — ongoing
ANSYS FEA	Finite element analysis — stress distribution	Report: NEPL/C-DISC/P-01/210425
PLAXIS 3D	Parametric study — multiple soil types	NIT Calicut MTech thesis
MatterLab, Wayanad	Field load testing — Wayanad soil conditions	Peak load: 667 kN/m²
GRIHA	Green Rating for Integrated Habitat Assessment	Certified
Patent	Product patent	Held by C-DISC Technologies Pvt. Ltd.

Four Target Audiences

The website must speak to four distinct buyer types. Each has a different primary fear and a different decision-making language. The website routes each visitor into a tailored experience — but all routes lead to the same consultation CTA.

Audience	Primary Fear	Decision Language	Key CTA
Civil Engineer / Architect	"Will this pass structural scrutiny?"	IS codes, FoS, load test data, NIT/IIT validation	Request a Structural Consultation
Homeowner (G+0 / G+1)	"Will my house be safe? Will it make a mess?"	Speed, cleanliness, disruption, cost	Check if PEN Works for Your Plot
Solar / EPC Developer	"Will this slow my commissioning timeline?"	MW, IRR, generation days, water logistics	Calculate Your MW Commissioning Advantage
Eco-Resort / Hospitality Builder	"Will this damage the ecology I'm selling?"	Zero excavation, tree roots, forest approvals, GRIHA	Plan Your Eco-Resort Foundation

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PART 1

Design Direction (To be Finalized yet)

Design Direction

This section describes the aesthetic vision for the website. Specific colour hex values, typography choices, and visual style are left to the design team to develop and propose. The direction below describes intent, feeling, and constraint — not prescription.

Aesthetic — Hybrid Dark / Warm

The website operates on two visual registers that alternate as the user scrolls:

Dark Sections	Warm Sections
Hero, key transition moments, and data/stats sections. Dark, dramatic, authoritative. Signals: DeepTech. Engineering precision. Confidence. These sections should feel like the inside of a control room or the depth of the earth.	All content, explanation, and story sections. Warm, earthy, organic. Signals: Biomimetic. Kerala-rooted. Human. These sections should feel like the inside of a well-built home — safe, grounded, trustworthy.

The transition between dark and warm sections is not just aesthetic — it IS the product story. Dark (below ground, raw, technical) transitions to warm (above ground, completed, liveable). The designer should feel free to use this transition as a storytelling device.

Colour Direction — For the Design Team

🎨 **DESIGN INTENT — Colour Philosophy**

– The brand has three anchors: earth, ecology, and engineering precision.

– Earth tones should dominate — the product comes from the soil and respects it.

– An ecological accent (green family) appears wherever sustainability or environmental benefit is communicated.

– A precision accent (cool tone — steel, slate, or blue family) appears for all technical data, IS codes, and engineering credentials.

– A warm highlight (gold or amber family) is used sparingly — for key proof numbers only: 2 hours, 667 kN/m², 2×. Never as a primary tone.

– No safety orange. No construction yellow. No generic grey. PEN is not a commodity.

– Dark backgrounds should feel like soil depth — not corporate dark mode.

– The designer should propose a palette that could be photographed in a Kerala forest and feel at home there.

Typography Direction — For the Design Team

🎨 **DESIGN INTENT — Typography Philosophy**

– Two fonts maximum: one display/headline font, one body/UI font.

– Headline font: should have warmth, authority, and a slight premium quality. Serif or refined sans-serif. Avoid generic web defaults.

– Body font: clean, highly readable at small sizes, works well on mobile. Humanist or geometric sans-serif.

– A monospace font (third typeface, optional) for hard data points only: load figures, IS codes, model numbers, dimensions.

– Monospace data points evoke engineering blueprints — they signal "this is measured, not estimated".

– Typography must be tested on mobile at 360px width — 50% of the target audience (site engineers and contractors) will be on a phone on a construction site.

Visual Style — Key References

These references describe the feeling the design should evoke — not the appearance to copy.

Reference	What to Take From It	What to Avoid
Apple product pages	Scroll-triggered reveals, generous white space, technical precision in layout	Stark minimalism — PEN needs warmth, not coldness
Dyson website	Engineering cross-sections shown elegantly, technical pride, material closeups	The grey/silver palette — PEN is earthy, not metallic
Kerala tourism photography	Laterite earth, forest light, coconut palms, monsoon imagery	Touristic clichés — this is a technology brand
Engineering blueprints	Grid paper, dimension lines, hatch marks, title blocks, annotation style	Literal blue/white — translate the feeling, not the colour
CarbonCure / GoliathTech	Construction tech credibility — professional but not intimidating	Generic construction site aesthetic — orange, yellow, grey

CTA Architecture — The Golden Rule

Every single section of the website must end with a call to action. No section is a dead end. CTA language escalates progressively as the visitor moves deeper into the page — from curiosity at the top to commitment at the bottom.

Stage	CTA Examples	Purpose
Curiosity	"See How It Works" / "Scroll to find out"	Move from hero into the technology story
Engagement	"Run the Numbers" / "See Your Soil Type" / "Watch the Installation"	Draw visitor into interactive tools
Trust	"Download the NIT Calicut Report" / "Read the Case Study"	Satisfy the skeptic with third-party evidence
Commitment	"Tell Us About Your Site" / "Request a Technical Assessment"	Primary conversion — qualified consultation lead

PART 2

Homepage Option — Animation First

Homepage Option — The Living Installation

This homepage leads with the PEN installation animation as the primary experience. The right side of the screen carries a persistent, scroll-driven line art animation that shows the PEN being installed in real time as the user reads. By the time the visitor reaches the final CTA, they have unconsciously watched a complete installation.

Layout Structure — Homepage Option A

Two-column layout for all sections:

- Left column: Text content, copy, interactive tools, CTAs
- Right column: Persistent scroll-anchored installation animation
- On mobile: Right column collapses to full-width animation panels between sections

The Living Right Rail — Condensed Version

The right column carries this animation from the top of the page to the bottom. Each scroll milestone triggers the next state. The animation is SVG line art — thin single-weight strokes, lightweight, scalable. No video. The style should feel like a precision engineering illustration coming to life.

▸ **SCROLL ANIMATION — Living Right Rail — Homepage A (Condensed)**

Step 1 Bare laterite earth. Soil texture visible in cross-section. Nothing there. Quiet.

Step 2 A ghost overlay shows a conventional concrete excavation — chaotic, destructive. It fades. Earth restores itself.

Step 3 The M50 precast concrete node appears on the surface. Faint glow or highlight to draw the eye.

Step 4 First GI pipe nail drives in at a batter angle. Ripple or friction lines radiate from the penetration point into the surrounding soil.

Step 5 Second, third, fourth nail follow in sequence. A force distribution halo pulses from each tip — showing skin friction engaging.

Step 6 Load arrows appear above the node pressing down. They split into four paths along the nails, dissipating into the soil as a warm diffuse glow. The load is held. Safe.

Step 7 A structural column rises from the node. Then beams. Then a completed structure.

Step 8 Full building visible above ground. The earth around the base: completely undisturbed. Animation rests — complete.

Left Column Content — Section by Section

Each section below corresponds to a scroll milestone in the right rail animation.

A01 — Hero

🎨 **DESIGN INTENT — Section Design Intent**

– Full-screen dark section. The question appears large — no product name, no logo initially.

– Four role-selector cards below the question. Minimal. Visitor taps their identity.

– Animation has not started yet — right rail shows bare earth only.

> ***"When was the last time a foundation surprised you?"***

No headline. No product pitch. Just the question. The visitor pauses. Then scrolls to find the answer.

Role Selector — Four Cards

- I am a Homeowner
- I am a Civil Engineer / Architect
- I am a Solar / EPC Developer
- I am an Eco-Resort / Hospitality Builder

Each card links to the relevant audience deep-page. But the homepage story continues regardless of selection.

> **CTA → Scroll to find out ↓**
> *Animated scroll indicator — no text CTA. Curiosity drives the scroll.*

A02 — The Problem

🎨 **DESIGN INTENT — Section Design Intent**

– Transition from dark to warm background — signals: we are moving from problem to possibility.

– Clock vs. Calendar split visual — full width. No headline needed. The visual IS the argument.

– "Before You Finish Reading This" counter — a live timer from page load. Subtle. Devastating.

Clock vs. Calendar Visual

Left: An analogue clock counting from 00:00. Right: A calendar with pages tearing off — Day 1, Day 2... Day 21. Both visuals reach their end simultaneously. Clock stops at 2 hours with a checkmark. Calendar stops at Day 21 with a neutral marker. No labels needed initially. Labels appear after a beat.

PEN Foundation	Conventional Foundation
2 hours. Zero curing. Load-bearing immediately.	21 days minimum. Curing mandatory. Waiting is the product.

"Before You Finish Reading This" Counter

> "You have been on this page for [live timer] seconds. In that time, a PEN Foundation was being installed. A conventional foundation has 20 days, 23 hours and 59 minutes to go." — Timer begins from page load. Runs live. Updates every second.

CTA → See How We Solved It →

Scroll trigger — moves to Technology section

A03 — Their Method / Ours

🎨 DESIGN INTENT — Section Design Intent

– Full-width interactive Before/After drag slider.

– Slider handle label: "Their Method / Ours" — not "Before / After".

– Left: Aerial photograph of a conventional foundation site — scarred, muddy, machinery tracks, excavated soil.

– Right: Same plot after PEN installation — grass intact, slight node indent, no tracks, no disruption.

– Three data points appear only after the visitor has dragged the slider at least once.

Data Points (appear post-interaction)

- 0 m^3 soil excavated
- 0 litres of water used in installation
- 0 trees disturbed — Black Langur Resort, Wayanad

CTA → Read the Wayanad Story →

Routes to Case Studies section

A04 — Nature Figured This Out First

🎨 DESIGN INTENT — Section Design Intent

– Emotional core of the page. Does not sell — it resonates.

– Root System Metaphor animation: tree above ground → cross-section reveals roots below → roots transform into GI pipe nails → node appears at trunk base → structure rises above.

– Full-width transition animation. Takes 6–8 seconds at normal scroll pace.

– Tagline appears over the completed image.

> ***"Nature figured this out. We engineered it."***

The Tree Distributes wind load through lateral root friction. Four primary roots at batter angles. No single point of failure.	**PEN Foundation** Distributes structural load through skin friction. Four battered GI pipe nails. No single point of failure.

Pre-Engineered Nail Foundation is not a metaphor. It is a direct biomimetic translation — the load transfer physics of a root system, expressed in steel and precast concrete. Validated by NIT Calicut. Governed by IS 2911 Part 4.

> **CTA → Understand the Engineering →**
> *Routes to Technology page*

A05 — The Weight Transfer Animation

PEN Foundation Load distributed via skin friction across four nail surfaces. Enhancement Factor: 2.0–2.6× — field-validated, NIT Calicut.	**Conventional Footing** Load concentrated on base area. Dependent on soil bearing capacity directly below the footing.

🎨 DESIGN INTENT — Section Design Intent

– Return to dark background — signals going underground, technical depth.

– 150 kN load appears above the node as a gold number. User watches it travel down through the system.

– Load splits into four paths along the nails — dissipates into soil as friction glow.

– Cut to conventional footing: same load pushes straight down on small base area. Stress concentrates.

– Final line appears: "One pushes down. One spreads out."

– No other copy needed in this animation moment.

Source: NIT Calicut Field Plate Load Tests · IS 2911 Part 4 · PLAXIS 3D Parametric Study

CTA → Download the NIT Calicut Validation Report →
Gated: name + email + organisation. Lead capture.

A06 — The Engineer's Sketchbook

🎨 **DESIGN INTENT — Section Design Intent**

– Background shifts to grid paper texture — light grid lines at low opacity.

– All illustrations in this section are hand-drawn engineering sketch style. Thin lines, annotation style.

– A toggle button allows visitor to switch between "Realistic Render" and "Blueprint View".

– Feels like someone who actually understands the engineering drew this — not a marketing team.

– A title block in the corner contains monospace technical data — like a real engineering drawing.

Annotated Diagram Content

- Node: "M50 Precast Concrete Node — 450mm × 450mm × 200mm"
- Nail: "GI Pipe — 32mm OD, 3.2mm wall, IS 1239"
- Tip: "Tungsten Carbide — Recycled Grade"
- Angle: "Batter Angle: 40–51° — Site-Adjusted"
- Depth: "Min. Embedment: 900mm — IS 2911 Part 4"
- Friction zone: "Skin Friction Zone" — pencil hatching to indicate the engaged soil volume

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MODEL          CD-PEN-32.3.2.1500
STANDARD       IS 2911 PART 4
FOS            2.5 (SINGLE TEST)  /  2.0 (DUAL TEST)
VALIDATED      NIT CALICUT / IIT KANPUR / MATTER LAB
CERTIFIED      GRIHA  /  PATENT HELD
```

CTA → Download Full Technical Specification →
Gated behind email — spec sheet PDF

A07 — The Time-Value Calculator

🎨 **DESIGN INTENT — Section Design Intent**

– Interactive calculator. Numbers update in real time as sliders move — no "Calculate" button.

– Three output cards: Days Saved (large prominent number), Interest Saved (rupee figure), CO_2 Avoided.

– A fourth output — "The Eureka Graph" — shows cumulative cash flow: Concrete project vs. PEN project. The gap between the two lines is shaded and labelled "Your Advantage".

– Below the calculator: email capture prompt — "Save your calculation."

Input Fields

- Project Type — Residential G+1 / Solar Ground Mount / Commercial / Eco-Resort / Industrial
- Project Scale — number of foundations, or sq ft, or MW
- Loan / Capital Interest Rate (%) — slider
- Daily Revenue or Rent Value ₹ — optional, for commercial projects

Calculation Logic

- Days Saved: 21 days (conventional curing) vs. 0 (PEN). Fixed.
- Interest Saved: Project loan value × daily interest rate × 21 days saved.
- CO_2 Avoided: Based on C-DISC Technologies internal LCA study. Source must be cited.
- Revenue Accelerated (optional): Days saved × daily revenue value input by user.

> **CTA → Get a Custom Project Assessment →**
> *Primary conversion — routes to consultation intake form*

A08 — What Goes Into the Ground

🎨 **DESIGN INTENT — Section Design Intent**

– Two columns build upward simultaneously — a stacking animation.

– Conventional column: cement bags, aggregate, water drums, shuttering wood, excavator icon. Pile grows tall and chaotic.

– PEN column: one node, four pipes. Pile stays tiny.

– A weighing scale at the bottom tilts hard toward Conventional for materials and CO_2.

– Data appears after animation completes.

Conventional Footing — Materials	**PEN Foundation — Materials**
Cement ~600kg · Aggregate ~1,200kg · Water ~300L · Shuttering timber · Excavator rental · Dewatering pump (where needed)	M50 Precast Node — ~48kg · Four GI Pipes — ~22kg · Fosroc Grout — ~4kg · Tungsten Carbide Tips · M12 Bolts

> **CTA → See the Full LCA Comparison →**
> *Routes to Sustainability section or downloads*

A09 — Case Studies

🎨 **DESIGN INTENT — Section Design Intent**

– Three case study cards. Format: Challenge → What PEN Did → Result.

– Each card has a download CTA for the engineering report.

– Photography: drone/site photos of each project. Real images only.

– Filter option: visitor can filter by project type.

> ### Devagiri Library, Calicut
> **CHALLENGE** Retrofitting under an existing occupied building. Zero tolerance for vibration or structural disruption. Existing flooring could not be broken.
>
> **RESULT** Installation completed without disrupting the structure above. Building remained operational throughout. Zero downtime.

> ### Black Langur Resort, Wayanad
> **CHALLENGE** Dense forest site. Live root systems of protected trees within close proximity of all foundation points. Forest department zero-excavation mandate.
>
> **RESULT** All foundations installed without disturbing a single root system. Forest department approval retained. Site completely undisturbed post-installation.

> ### Rehabilitation Housing — Wayanad
> **CHALLENGE** Post-landslide slope. Unstable laterite soil. Conventional foundations impossible — ground too disturbed for excavation.
>
> **RESULT** PEN foundations installed on active slope within 72 hours of site clearance. Immediate load-bearing — no curing wait on unstable terrain.

> **CTA → Download Engineering Reports →**
>
> *Gated: name + email + organisation*

A10 — Soil Type Selector

🎨 DESIGN INTENT — Section Design Intent

– Five soil type swatches shown as visual texture tiles — not a text dropdown.

– Visitor taps/clicks their soil type.

– Right rail (or below on mobile) updates to show how the PEN nail engages with that specific soil.

– Case study below updates to show the most relevant real project for that soil condition.

– Load capacity range updates for that soil condition.

– Any special installation notes appear (e.g., pilot bore for rocky sites).

– Soil type selection is pre-filled into the consultation form.

Soil Type Options

- Red Laterite — common across Kerala and South India
- Black Cotton — expansive clay — Maharashtra, Gujarat, Deccan
- Sandy — coastal and river sites

- Rocky — Western Ghats, hill sites
- Waterlogged / High Water Table — coastal lowlands, flood plains

> **CTA → Talk to a PEN Engineer About Your Soil →**
> *Consultation form — soil type pre-filled from selection*

A11 — The Site Office

🎨 DESIGN INTENT — Section Design Intent

– Visual: A hyper-realistic illustration of a site engineer's desk.

– Items visible: CAD foundation layout, physical PEN spec sheet, calculator, phone with missed calls ("Structural Consultant: 3"), coffee cup ring, GRIHA certificate pinned up, soil investigation report open.

– Every item is hoverable — tooltip appears linking to real PEN resources.

– This section is for contractors and site engineers. When they see it, they recognise their world.

– All data on the desk is real PEN data — not placeholder.

Hover Interactions

- Spec sheet → "Download the full technical specification"
- Calculator → "Run your own numbers — open the Time-Value Calculator"
- CAD drawing → "Download DWG/PDF of a standard PEN foundation layout"
- Soil report → "Understand what soil data you need before specifying PEN"
- GRIHA certificate → "What GRIHA certification means for your project"

> **CTA → Download the Contractor Specification Pack →**
> *Full pack: spec sheet + layout drawings + IS code compliance note*

A12 — The Final Close

🎨 DESIGN INTENT — Section Design Intent

– Warm, confident close. Not aggressive. The visitor has been through the full conviction journey — this is the handshake.

– Background: brand primary (designer to propose — warm, prominent, high contrast with white text).

– Large headline. One-line subtext. One primary CTA. One secondary link.

– Below: short consultation intake form embedded directly on the page.

> ***"Stop Pouring. Start Nailing."***

Tell us about your site. We'll tell you if PEN is right for it.

Consultation Intake Form

- Name
- Organisation / Company
- Project Type — dropdown
- Location — city or district
- Soil type (if known) — optional, pre-filled from Soil Selector if used
- Timeline — Immediate / 3 months / 6 months / Exploring
- How did you hear about PEN? — optional

CTA → Request a Technical Site Assessment →
Primary conversion. Submit = lead registered in CRM.

CTA → Or call us directly →
Phone number + WhatsApp link

PART 3

Technology Page

The Technology page is the engineering depth page. It uses the full, detailed version of the Living Right Rail animation — ten precise scroll states that walk through every component and every installation step. The left side provides detailed technical explanation. The right side shows it happening.

This page is the primary destination from the homepage "Understand the Engineering" CTA. It is also where engineers and architects will arrive from organic search.

Right Rail — Full Version (10 States)

▸ **SCROLL ANIMATION — Living Right Rail — Technology Page — Full Version**

Step 1 GROUND CROSS-SECTION: Laterite soil strata labelled in monospace. Topsoil 0–150mm / Laterite 150–600mm / Clay 600–1200mm. Nothing else present.

Step 2 NODE PLACEMENT: M50 precast node settles onto the surface. Dimensions annotated: 450 × 450 × 200mm.

Step 3 PILOT BORE: Thin borehole opens at each nail position. 40–51° angle guide lines appear.

Step 4 NAIL 1: First GI pipe drives in — North-West batter angle. Depth counter runs: 0mm → 900mm → 1200mm.

Step 5 NAIL 2, 3, 4: Remaining three nails follow in sequence. Each with live depth counter.

Step 6 GROUTING: Fosroc Conbextra GP2 grout flows down each pipe. Annular space fills. Colour change indicates grout reach.

Step 7 FORCE DISTRIBUTION: 150kN load applies above. Splits into four paths along nails. Skin friction glow radiates from each nail surface into surrounding soil.

Step 8 BOLT ASSEMBLY: M12 Grade 4.6 galvanised bolts secure the superstructure connection point.

Step 9 LOAD READY: Green confirmation marker. "Load-bearing from this moment." Install time shown: ~1hr 47min. No curing.

Step 10 COMPLETE: Column rises. Structure above. Earth around the base: completely undisturbed.

Left Side — Content Sections

T01 — The Mechanism

How PEN transfers load. Written for a structural engineer — technically accurate, no dumbing down, no unnecessary jargon. Explains the skin friction mechanism, the stressed soil cone concept, and why batter angles matter.

> **CTA → Download the Full Technical Report →**
> *Gated by email*

T02 — The Materials

Every component explained with material grade and source standard:

- GI Pipe — 32mm OD, 3.2mm wall, IS 1239 compliant, hot-dip galvanised
- Tungsten Carbide Tip — recycled Kennametal/Sandvik grade, bonded to pipe end
- M50 Node — precast, 28-day cured before dispatch from Factory F1
- Grout — Fosroc Conbextra GP2 micro-concrete, non-shrink
- Bolts — M12 Grade 4.6 galvanised, IS 5624 compliant

> **CTA → Download Material Specification Sheet →**
> *Gated by email*

T03 — The Validation Timeline

A horizontal timeline of all validation milestones — in chronological order. Each milestone card shows institution, what was tested, methodology, and key output. Cards are downloadable.

> **CTA → Download All Validation Reports →**
> *Gated: name + email + organisation*

T04 — IS Code Compliance

Full explanation of IS 2911 Part 4 governance. Why IS 6403 (shallow footings) does not apply. Factor of Safety derivation. This section directly addresses the most common objection from structural engineers.

> **CTA → Talk to a Structural Engineer — Book a Technical Call →**
> *Direct consultation with engineering team*

PART 4

Audience Deep Pages

Four dedicated pages — one per audience. Each page is a curated version of the PEN story told in the language of that specific visitor. Same brand, same evidence — different emphasis, different copy register, different primary CTA. Visitors arrive from the role selector on the homepage.

Page A — Civil Engineer / Architect

PRIMARY FEAR: "WILL THIS PASS STRUCTURAL SCRUTINY?"

Tone

Peer-to-peer. Technical without condescension. Every claim cited. Every number sourced. The Engineer's Sketchbook aesthetic dominates.

Content Emphasis

- IS 2911 Part 4 compliance — full explanation with code references
- Factor of Safety derivation — 2.5 (one test) / 2.0 (two or more tests)
- PLAXIS 3D parametric study — soil type variables and outputs

- The Objection Wall — common structural engineer objections answered with IS code citations (interactive sticky note UI)
- BIM/CAD downloads — DWG foundation layout, specification PDF
- NIT Calicut and IIT Kanpur academic credentials — prominently displayed

> **CTA → Request a Structural Consultation →**
> *Direct line to engineering team — not sales*

Page B — Homeowner (G+0 / G+1)

PRIMARY FEAR: "WILL MY HOUSE BE SAFE? WILL IT MAKE A MESS?"

Tone

Warm, reassuring, human. Minimal technical jargon. The story is: done in a day, no mess, your plot untouched. Visuals of clean residential sites.

Content Emphasis

- The "2 hours" timeline — shown as a comparison to a homeowner's typical day
- Before/After slider — applied to a residential plot, not a commercial site
- Cost context: G+1 home requires approximately 16–20 PEN units. Pricing on request.
- GRIHA certification in plain language — what it means for their home and resale value
- Testimonials — real homeowner quotes, in both English and Malayalam
- FAQ section: Can I build on my plot? What soil do I have? Do I need a soil test?

> **CTA → Check If PEN Works for Your Plot →**
> *Soil selector + consultation form*

Page C — Solar / EPC Developer

PRIMARY FEAR: "WILL THIS SLOW MY MW COMMISSIONING TIMELINE?"

Tone

Data-first. IRR-focused. Speak in MW, generation days, PPA timelines, water logistics. Time-Value Calculator is front and centre, pre-configured for solar projects.

Content Emphasis

- Zero water — no water trucking to remote sites. Fully dry process.
- Immediate load-bearing — module racking begins same day as foundation installation
- Solar-specific calculator — days of generation revenue gained by early commissioning
- Remote terrain capability — rocky slopes, sandy soil, no road access needed
- Time-lapse comparison visual — 1MW installation: PEN vs. conventional

> **CTA → Calculate Your MW Commissioning Advantage →**
> *Solar-configured Time-Value Calculator → consultation form*

Page D — Eco-Resort / Hospitality Builder

PRIMARY FEAR: "WILL THIS DAMAGE THE ECOLOGY I'M SELLING?"

Tone

Nature-first. Ecological. Premium. Photography-led. Kerala forests, Western Ghats terrain, resort settings. Root System Metaphor animation lives here in its full form.

Content Emphasis

- Zero excavation — no tree roots disturbed, no soil structure broken, no slope destabilised
- Black Langur Resort, Wayanad — primary case study, with drone photography and contractor quote
- Forest department clearance — PEN's zero-excavation process simplifies approval
- GRIHA certification — green building credential for sustainability positioning and LEED points
- The Carbon Shadow visual — show the environmental footprint difference at resort scale
- The invisibility of the foundation — premium positioning: "your guests will never see it"

> **CTA → Plan Your Eco-Resort Foundation →**
> *Consultation form — eco-resort category pre-selected*

PART 5

Technical Requirements

Technical Requirements

Recommended Stack

These are recommendations — the development team should propose their preferred stack and justify any deviations.

Layer	Recommended	Reason
Hosting	Vercel or AWS Amplify	Global edge caching — fast load in India and target markets
Forms / CRM	Zoho CRM	Consultation intake leads go directly to CRM pipeline

Analytics	Zoho or Google Analytics	Scroll depth, heatmaps, calculator engagement tracking

Performance Requirements

- Core Web Vitals: LCP under 2.5s, CLS under 0.1, FID under 100ms — on mobile
- Page load under 3s on a 4G connection in India
- All animations must be GPU-accelerated (CSS transform / will-change) — no layout-triggering animations
- Living Right Rail must be scrub-safe — animation state is correct at any scroll speed, including fast scroll
- Mobile fallback for right rail: full-width animation panels between sections (no two-column layout on mobile)
- All interactive tools (calculator, slider, soil selector) must be fully functional on touch devices

SEO Requirements

- Server-side rendering for all content pages — no client-side-only rendering for indexable content
- Structured data markup: Organisation, Product, FAQPage schemas
- All downloadable PDFs must have text-layer content indexed via page description
- Page titles and meta descriptions to be written by founder before build — not auto-generated
- Image alt text: all product and site photography must have descriptive alt text, provided by founder

Lead Capture & Gating

Two tiers of gated content:

Tier 1 — Light Gate Name + Email only. Unlocks: Case study PDFs, technical specification sheet, material spec sheet. Visitor stays on page.	**Tier 2 — Full Gate** Name + Email + Organisation + Project Type. Unlocks: NIT Calicut validation report, PLAXIS 3D study, full engineering pack. Triggers sales notification.

- All lead data goes to CRM immediately on form submit
- Auto-response email sent within 2 minutes of Tier 1 or Tier 2 submission
- Consultation intake form (Section A12/B09) is not gated — it IS the conversion event

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PART 6

Phase 2 — Features Not in Scope

Phase 2 Features — Not in Scope for This Build

The following features are deliberately excluded from the Phase 1 website. They are documented here for continuity only — so context is not lost when Phase 2 begins. Do not include these in the Phase 1 scope, quote, or timeline.

Feature	Why Not Now	What Triggers Phase 2 Build
Manufacturing / Franchise Partner Portal	No manufacturing partners in Year 1. Production-constrained.	First 3 manufacturing partners onboarded
Gamified Contractor Loyalty Programme	Requires critical mass of certified installers.	25+ certified installers active
AR Site Visualisation	High dev cost. Low conversion lift at this stage.	Post BMTPC certification — national scale
Find a Certified Installer Map	No public installer network yet.	Year 2 GTM — post-seed round
Digital Product Passport (QR Trace)	Requires manufacturing database integration.	Factory QC system built out

End of Brief

PEN Foundation · C-DISC Technologies Pvt. Ltd.